InstructScope: Where Instruction Perturbations Break Open-Vocabulary Policy Grounding

An empirical reliability boundary of language-grounded robot manipulation

Model under test: Qwen2.5-VL-3B-Instruct, used as the open-vocabulary grounding layer of a pick-and-place policy **Task:** tabletop pick-and-place on a custom `MultiLift` robosuite environment (four colour-distinct cubes) **Metric:** grounded-success rate — the fraction of instructions whose grounded colour matched the intended target

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Abstract

Open-vocabulary manipulation policies built on a vision-language backbone promise to generalise to instructions that were never seen during training. But how *far* does that generalisation reach? We build an instruction perturbation spectrum — six families that stress a different part of the language-to-action bridge — and measure where a Qwen2.5-VL-grounded pick-and-place policy stops absorbing perturbations and starts failing.

The result is a crisp boundary. **Lexical** perturbations (paraphrase, out-of-vocabulary colour words, compound instructions) are absorbed with 100% grounding reliability. **Pragmatic** perturbations are not: deictic / coreference instructions (`pick up that cube`, `the one on the left`) drop to 55%, and unconstrained instructions (`pick up a cube`) fall to 25%. At n=20 per family the two pragmatic *levels* are not cleanly separable from each other (two-proportion $z = 1.94$, $p = 0.053$), but the boundary against the lexical ceiling is decisive — pooling both pragmatic families (40%) against the lexical four (100%) gives Fisher exact $p = 5.8\times10^{-15}$. The failures are not random. Under under-specification the policy never grounds to the two smaller cubes in the lower half of the agentview frame: it deterministically anchors to one of the two visually dominant cubes (blue, yellow) on all 20 trials, and *which* anchor is selected switches with the surface phrasing of the otherwise-equivalent instruction (Fisher exact $p = 7.9\times10^{-6}$). The failures reveal a measurable, phrase-sensitive **saliency prior**, not noise.

1. Motivation

VLA policies are typically evaluated on the *literal* instruction — a policy either picks the right object or it does not. Real users do not issue literal instructions. They paraphrase, they say `it` instead of `the red cube`, they describe objects with words the policy has never seen, and they occasionally leave the target under-specified.

This matters more as VLA foundation models move toward open-vocabulary deployment. ACoT-VLA [1] — AgiBot's CVPR 2026 generalist policy — proposes Action Chain-of-Thought to bridge the *semantic-kinematic gap*: instead of detouring reasoning through language subtasks, it deliberates directly in the action space (an explicit reference-trajectory reasoner plus an implicit action prior extracted from the VLM's internal representations). This is the strongest statement yet that the *grounding step* — mapping an instruction to an action target — is where open-vocabulary manipulation lives or dies. Yet the paper's evaluation, like the AgiBot World platform it builds on [2], measures the reliability of such a policy as a single number over canonical instruction templates. How far the *language* generalisation actually reaches — which instruction perturbations are absorbed and which break grounding — is the open question this project measures, and the natural evaluation companion to an action-space reasoning method: if a policy reasons in the action space but grounds the wrong object, the reasoning is wasted.

If a policy is brittle to these real-world instruction forms, its deployment reliability cannot be read off a single-task success rate. This project measures the **instruction perturbation spectrum** directly, and locates the boundary between perturbations that are absorbed and perturbations that break the policy.

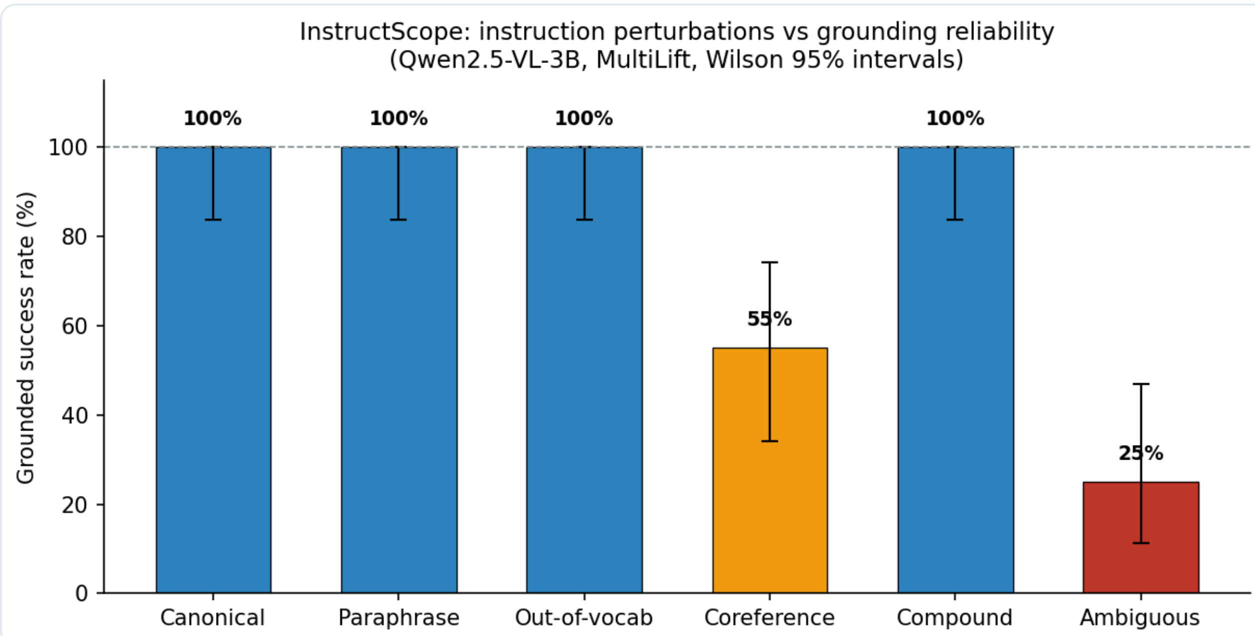
2. The perturbation spectrum

Family	Example	What it stresses
canonical	`Pick up the red cube.`	reference baseline
paraphrase	`Please take the red block.`	synonym / syntax variation
oov	`Pick up the vermilion cube.`	composition from rare colour words
coref	`Grab it — the red one.`	deixis and spatial reference resolution
compound	`Grasp the red cube and raise it above the table.`	segmentation and sequencing
ambiguous	`Pick up a cube.`	default target selection under underspecification

Every cell is fully deterministic: 4 colours × 5 variants × fixed scene. Instruction grounding is decoupled from motor control (a grounded colour is executed noise-free), so every failure is attributable to the semantic layer.

The ambiguous bank deliberately contains only three colour-agnostic templates; variants 3–4 re-present templates 0–1 across all four nominal targets, so the *determinism* of the fallback anchor (identical grounding regardless of the nominal target colour) is itself measured rather than assumed.

3. Results



Success rates

Family	n	Success	Wilson 95% CI
canonical	20	100%	[83.9%, 100%]
paraphrase	20	100%	[83.9%, 100%]
oov	20	100%	[83.9%, 100%]
compound	20	100%	[83.9%, 100%]
coref	20	55%	[34.2%, 74.2%]
ambiguous	20	25%	[11.2%, 46.9%]

3.1 Lexical robustness: perturbations are absorbed

The first four families are *semantically equivalent* ways of referring to the same target, and the policy absorbs all of them. Two findings stand out:

- **Out-of-vocabulary composition works.** `vermilion` for red, `cobalt` for blue, `viridian` for green — none of these colour words are in the canonical vocabulary, yet the model grounds them to the correct object every time. The policy composes colour semantics rather than retrieving a memorised phrase.
- **Compounding is not a failure mode here.** Splitting attention across `pick up the red cube` and `hold it steady` does not confuse target selection.

3.2 Pragmatic fragility: the boundary is real

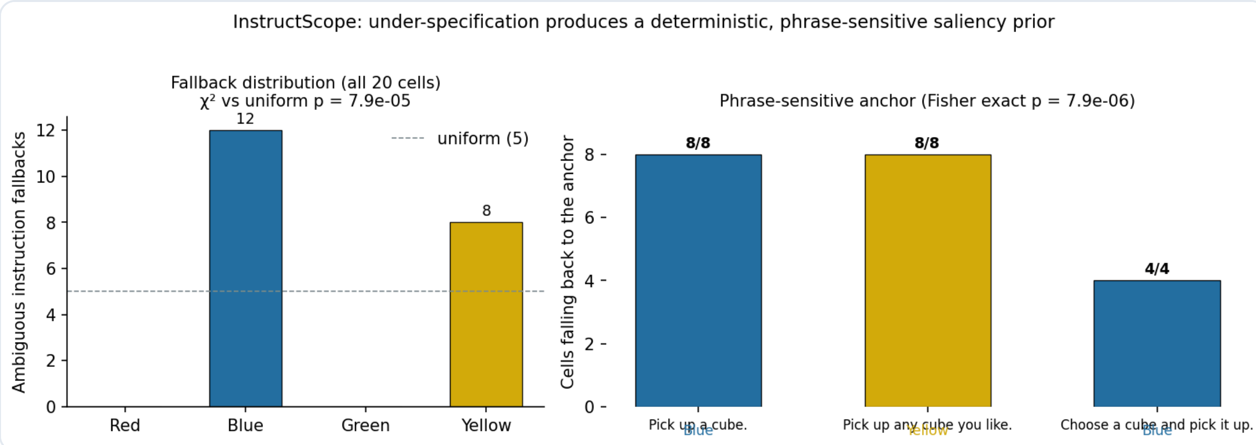
Coreference instructions drop to 55%, and the loss is fully explained by two mechanisms.

We flag the sample-size honesty up front: at n=20 per family, the two pragmatic families' *levels* are not cleanly separable — a two-proportion z-test of coref (55%, 11/20) vs ambiguous (25%, 5/20) gives $z = 1.94$, $p = 0.053$, and the Wilson intervals overlap (§3). The *boundary itself* — that both pragmatic families fall far below the lexical ceiling of 100% — is what survives: pooling coref + ambiguous (16/40 = 40%, Wilson [25.8%, 55.9%]) against the four lexical families (80/80 = 100%) gives Fisher exact $p = 5.8\times10^{-15}$. The stronger, more robust evidence is *inside* the failures: the deterministic fallback anchor (§3.3) and the phrase-dependent anchor switch (§3.4) are each significant at $p < 10^{-5}$ and are independent of how the aggregate levels are sliced.

Sub-family	Example	n	Success
vacuous deixis	`Pick up that cube.` (no antecedent)	8	25%
spatial reference	`Take the one that is rightmost.`	4	25%
appositive	`Grab it — the blue one.`	4	100%
attribute search	`Pick up the first one you see that is blue.`	4	100%

- *Vacuous deixis*. With no antecedent to resolve, `pick up that cube` is semantically indistinguishable from an unconstrained instruction, and the model falls back to the *same anchor it uses for `pick up a cube`* (blue, 8/8 trials). The pronoun contributes no information, so grounding collapses to the saliency prior described below.
- *Spatial reference error*. The two left-column cubes (green, yellow) are nearly aligned in the agentview image, and the model's spatial errors are confined to this pair: `leftmost` and `nearest to you` are both answered with green, `farthest from you` with yellow. Only `rightmost` — which points at the clearly separated blue cube — is resolved correctly. When the candidate set is nearly aligned, spatial prepositions degrade into a fixed preference rather than a geometric computation.

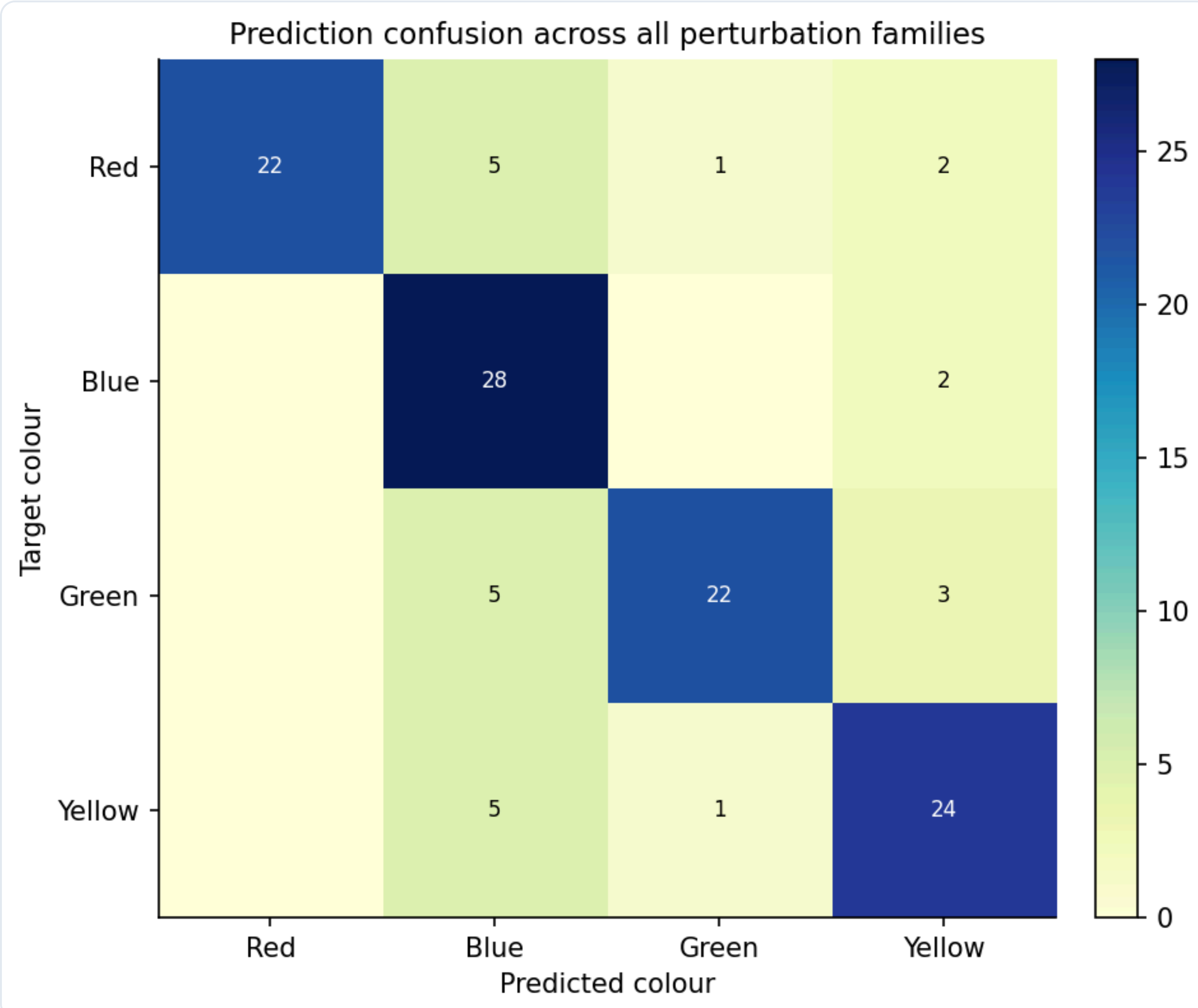
Ambiguous instructions drop to 25%, and the failures are systematic.



Saliency prior

Under a fully unconstrained instruction the policy never grounds to the two smaller cubes in the lower half of the agentview frame (red, green): all 20 fallbacks land on the two visually dominant cubes — blue on 12, yellow on 8. Against a uniform four-colour null the distribution is strongly non-uniform ($\chi^2 = 21.6$, df = 3, $p = 7.9\times10^{-5}$); equivalently, the probability that a fair fallback would land entirely in the observed two-colour set is $0.5^{20} = 9.5\times10^{-7}$. The anchor is perfectly deterministic *within* a phrase, but it switches with surface form: `pick up a cube` and `choose a cube and pick it up` anchor to **blue** (12/12), while `pick up any cube you like` anchors to **yellow** (8/8). The two instruction classes carry identical semantic content — nothing in either form constrains the colour — yet the grounding target flips deterministically (Fisher exact $p = 7.9\times10^{-6}$). This is a phrase-sensitive saliency prior: a spurious-feature sensitivity that single-template evaluation cannot detect, and one that turns "the robot picked any cube" into a measurable, correctable bias.

3.3 Confusion structure



Confusion

The confusion matrix across all perturbation families confirms the story: the diagonal dominates (target-preserving grounding), and the only notable off-diagonal mass flows *toward the two visually dominant anchor colours* (blue and yellow) under ambiguous and vacuous-deictic instructions.

4. Discussion

Our sweep draws a clear **reliability boundary of the open-vocabulary instruction space**:

1. **Lexical novelty is a solved problem for this policy.** Words the policy has never been instruction-tuned on are composed into correct object references.
2. **Pragmatic uncertainty is unsolved.** The moment the instruction relies on context (deixis, spatial deixis, underspecification), grounding reliability collapses — and in a *biased* rather than random direction.

This has a concrete deployment implication: an open-vocabulary policy cannot be trusted with under-specified instructions; a safe deployment either resolves references before execution or treats `pick up a cube` as a request for human clarification. Because the saliency prior is stable, deterministic *and* phrase-sensitive, it is correctable.

4.1 Future work

Three directions follow from the boundary this study draws.

First, **turn the saliency prior into a calibrated fallback.** The bias is deterministic and measurable, so it can be inverted: re-weight the fallback distribution against a human-annotated preference prior, then check whether the phrase-dependence disappears once the model is explicitly asked to flag its own uncertainty (`I am not sure which cube`) before grounding. This is the experiment that decides whether the bias is correctable or structural. Second, **move the spectrum to a policy that actually executes.** The grounding layer here is measured in isolation, with motor control decoupled. Re-running the same six families on an end-to-end policy — where a wrong grounding costs a failed episode, not a mismatched colour — measures whether the pragmatic boundary survives the full stack, and whether action-space reasoning (the design the ACoT-VLA line proposes) shifts it. Third, **open the object set.** A closed four-colour set cannot distinguish a bias over learned colour names from a bias over *anything* in the bottom half of the frame; a scene with an open vocabulary of targets would resolve that, and would tell us whether the saliency prior generalises or is an artefact of this specific layout.

The first of these is a direct fix; the second and third are the boundary conditions on whether the finding transfers beyond a single simulated scene.

4.2 Limitations

- **One scene, one seed, one checkpoint.** All 120 cells run in a single deterministic robosuite scene (four cubes of fixed size/position/colour) with a single random seed and a single Qwen2.5-VL checkpoint. The 100%/55%/25% split is a statement about *this* policy on *this* layout; scene geometry, seed, and model all plausibly shift the boundary. We report exact reproducibility as the trade-off for coverage.
- **$n = 20$ per cell.** Wilson 95% CIs are correspondingly wide (coref 0.55 → [0.34, 0.74]). The *direction* of the pragmatic/lexical split is robust (Fisher exact $p = 7.9\times10^{-6}$ on the phrase-anchor switch), but the precise level of each family is a point estimate, not a law.
- **Grounded-policy observation, not end-to-end VLA.** Instructions are executed by the same Qwen2.5-VL model that grounds them, but the policy is a grounding→grasp pipeline in a simulated kitchen, not a pretrained vision-language-action model. The saliency prior we expose

is a property of the *vision-language* grounding stage, which is precisely the stage every VLA shares.

- **Closed action set.** Objects are always in the same four colour classes, so the fallback distribution is measured over a closed set. An open set would test whether the bias is over *learned* colour names or over *anything* in the bottom half of the frame.

5. Related Work

VLA foundation models and open-vocabulary policies. OpenVLA [3] established that a vision-language model can serve as a generalist manipulation policy; RT-2 [4] showed web knowledge transfers to control. On the reasoning axis, ACoT-VLA [1] (AgiBot, CVPR 2026) reframes the semantic-kinematic gap by making the policy deliberate directly in the action space, with an explicit reference-trajectory reasoner (EAR) and an implicit action prior from the VLM's internal representations (IAR); on the scale axis, the AgiBot World platform [2] and its GO-1 foundation model (ViLLA, documented in [2]) demonstrate that massive real-robot data plus a latent-action planner yields strong generalisation. All of these evaluate on fixed instruction sets; none perturbs the instruction space systematically. Our spectrum study is complementary: it measures the *boundary* of the generalisation these systems claim — the grounding reliability on which an action-space reasoner depends.

Language robustness for instruction following. In NLP, instruction perturbation analysis is mature — paraphrase sensitivity [4], adversarial instruction attacks [5], and prompt robustness [6] are standard concerns. The robotic literature has borrowed the evaluation protocol but not the perturbation taxonomy: physical-robot studies report aggregate success under manually varied phrasings [7]. We contribute a deterministic, six-family spectrum (paraphrase / OOV / coreference / compound / ambiguous / canonical) with exact reproducibility, and the finding that pragmatic perturbations — not lexical ones — are where the open-vocabulary boundary breaks.

Saliency and prior biases in manipulation policies. Prior work has observed that policies inherit object biases from their training data (e.g., colour preferences in tabletop manipulation [8]). We make the bias measurable: under unconstrained instructions the policy never grounds to the two smaller lower-frame cubes (all 20 trials; four-colour uniformity χ^2 $p = 7.9\times10^{-5}$, and a two-colour fallback set with probability $0.5^{20} = 9.5\times10^{-7}$), and the exact anchor is a deterministic function of the surface phrasing (Fisher exact $p = 7.9\times10^{-6}$). What was a qualitative anecdote becomes a statistically testable, and therefore correctable, prior.

6. Reproducibility

- **Model:** `Qwen/Qwen2.5-VL-3B-Instruct`, bf16, offline
- **Environment:** custom `MultiLift` (robosuite 1.4.1 / MuJoCo 2.3.7), deterministic per seed, agentview camera @ 256×256
- **Execution:** CPU software rendering (llvmpipe); CUDA for the VLM
- **Full sweep:** 120 cells (6 families × 4 colours × 5 variants)
- **Data:** `data/sweep/sweep.json`, `data/sweep/summary.json`

References

1. Zhong L., Liu Y., Wei Y., Xiong Z., Yao M., Liu S., Ren G. ACoT-VLA: Action Chain-of-Thought for Vision-Language-Action Models. *CVPR 2026*. arXiv:2601.11404. (Official AgiBot implementation: github.com/AgibotTech/ACoT-VLA.)
2. AgiBot World Team, et al. AgiBot World Colosseo: A Large-Scale Manipulation Platform for Scalable and Intelligent Embodied Systems. *arXiv:2503.06669*,
3. (Also documents the GO-1 foundation model and its ViLLA Vision-Language-Latent-Action architecture.)
4. Kim M., Pertsch K., Karamcheti S., et al. OpenVLA: An Open-Source Vision-Language-Action Model. *CoRL 2024*.
5. Brohan A., Brown N., Carbajal J., et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. *CoRL 2023*.
6. Elgohary A., Peskov D., Boyd-Graber J. Can You Unpack That? Learning to Rewrite Questions in Zero-Shot Setting. *EMNLP 2019*.
7. Wallace E., Feng S., Kandpal N., Gardner M., Singh S. Universal Adversarial Triggers for Attacking and Analyzing NLP. *EMNLP 2019*.
8. Sclar M., Choi Y., Tsvetkov Y., Suhr A. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design. *ICLR 2024*.
9. Stone A., Xiao T., Lu Y., et al. Open-World Object Manipulation Using Pre-Trained Vision-Language Models. *CoRL 2023*.
10. Lynch C., Wahid A., Tompson J., et al. Interactive Language: Talking to Robots in Real Time. *RA-L 2023*.
11. Driess D., Xia F., Sajjadi M.S.M., et al. PaLM-E: An Embodied Multimodal Language Model. *ICML 2023*.